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188541

Level-4, 2nd Sem / DVOC
(Medical Imaging Technology)
Subject : Physics & Technology in imaging

Time : 2 Hrs.

M.M. : 50

SECTION-A

Note: Multiple choice questions. All questions are compulsory (5x1=5)

- Q.1 What is the primary benefit of using rotating anode X-Ray tubes rather than stationary anode tubes?
- a) Decreases radiation exposure
 - b) Reduced voltage ripple
 - c) Ability to use higher kVp
 - d) Ability to use higher mAs
- Q.2 Increased density on X-Ray film denotes
- a) A black shadow
 - b) White shadow
 - c) Increased radio lucency
 - d) None of the above
- Q.3 What is the source of X-Ray photon in tube
- a) Cathode
 - b) Rotor
 - c) Filament
 - d) Anode

- Q.4 The cathode end of X-Ray tube possesses a _____ charge
- a) Positive b) Negative
c) Neutral d) None of above
- Q.5 A device that converts mechanical energy to electrical energy called
- a) Generator b) Motor
c) Induction Motor d) Transformer

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

- Q.6 Expand KVp.
Q.7 Heat Unit.
Q.8 Energy flux density.
Q.9 Define pH.
Q.10 Radiology.

SECTION-C

Note: Very Short answer type questions. Attempt any six questions out of eight questions. (6x5=30)

- Q.11 Photoelectric Effect and its importance in diagnostic radiology.

- Q.12 Explain Three regions of operation of Gas filled detectors.
Q.13 Explain grids and types of grids.
Q.14 Explain Thermionic emission.
Q.15 Explain Cathode ray tube.
Q.16 Describe a Geiger Muller counter and explain its working.
Q.17 Explain the basic concept and working principle of scintillation counter.
Q.18 What are the biological effects of ionization radiation.

SECTION-D

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

- Q.19 What are the major components of an X-Ray tube? Explain how X-Rays are produced.
Q.20 Explain the factors affecting the X-Ray beam quality.